Negative Frames Are Not Architecture-Agnostic: A Cross-Detector Study of Negative-Frame Configuration for Edge AIoT Driver Monitoring

1st Oumar Mamoun Ibrahim
Dept. of Computer Engineering
University of Sharjah
Sharjah, United Arab Emirates
U22200741@sharjah.ac.ae
ORCID 0009-0008-0312-1605

2nd Mohamad Khairi bin Ishak
Dept. of Computer Engineering
University of Sharjah
Sharjah, United Arab Emirates
mishak@sharjah.ac.ae
ORCID 0000-0002-3554-0061

***Abstract*—Edge AIoT object detectors, such as in-cabin driver-monitoring systems, continuously process video streams in which background-only (negative) frames dominate the natural operational distribution (approximately 81% negative prevalence in our dataset). While incorporating negative frames during training mitigates false positives, prior studies tune the negative-to-positive proportion in isolation for individual detector architectures, leaving open whether an optimally tuned configuration transfers across architecturally distinct models. This paper presents the first controlled, cross-architecture empirical investigation of negative-frame ratio sensitivity and curation quality across three lightweight edge detectors: YOLO11n (anchor-free CNN), YOLO26n (next-generation CNN), and D-FINE-N (transformer-based real-time DETR). Evaluating five nested ratio configurations (0% to 80% negative frames) and a hard-negative mining strategy at matched dataset size on a 15,723-frame driver-monitoring corpus, we investigate whether the performance-maximizing ratio generalizes across detector lineages. We further translate false-positive rates per 1,000 frames into estimated nuisance alert rates per hour, demonstrating that training-data curation directly governs edge deployment usability independently of model inference latency.**

***Index Terms*—AIoT, edge intelligence, driver monitoring system, object detection, negative frames, hard-negative mining, YOLO, D-FINE, false-positive rate, edge deployment**

I. Introduction

Object detectors deployed in AIoT systems—such as in-cabin driver monitoring, smart surveillance, and industrial defect inspection—operate under a condition that standard computer-vision benchmarks rarely reflect: the vast majority of frames encountered at inference time contain *no* target object. A driver-monitoring camera streams continuously for hours per trip, yet the events it must flag (distraction, drowsiness, unsafe behavior) occupy only a small fraction of that stream. This asymmetry means a detector's behavior on background-only frames—rather than its accuracy on object-dense benchmark images—ultimately determines whether the system is usable in the field. A model prone to false positives on ordinary footage produces nuisance alerts that erode user trust, waste edge compute cycles, and, in bandwidth- or battery-constrained AIoT deployments, trigger unnecessary uplink transmissions or actuator responses.

The standard mitigation is to include negative (background-only) frames during training, and prior work confirms this lever is powerful. Reported optimal negative-frame proportions range from single digits [8] to over 60% of the training set in safety-critical driving contexts [6], with performance following a non-monotonic, inverted-U relationship: too few negatives leave false positives unaddressed, while too many suppress recall by pushing the detector toward over-conservative predictions [5]. However, every empirical study we are aware of tunes this parameter for a *single* detector architecture in isolation. It remains unknown whether a negative-frame configuration—including ratio or the choice between random and hard-mined negatives—tuned for one architecture transfers to another, or whether the optimum is itself architecture-dependent.

This is not a minor gap: AIoT practitioners routinely choose among CNN-based, anchor-free, and transformer-based lightweight detectors under a shared, fixed data-curation budget. If the answer to "how many negative frames do I need" changes with the architecture choice, then architecture selection and data-curation decisions cannot be made independently—as much current practice implicitly assumes.

This paper addresses that gap through a controlled, cross-architecture study of negative-frame configuration on a real-world, safety-critical AIoT dataset. Using an in-cabin driver-monitoring corpus with a naturally occurring 19:81 positive-to-negative frame ratio, we train three lightweight detectors spanning distinct design paradigms (YOLO11n, YOLO26n, and D-FINE-N) under systematically varied negative-frame ratios and sampling strategies, holding all other training factors fixed. We ask:

- **RQ1:** Does the negative-frame ratio that maximizes detection performance differ across detector architectures, or does a single configuration generalize?

This work was supported by the University of Sharjah. Code and configurations are available at `https://github.com/IamOumarIbrahim/ieee-aiot`.

- **RQ2:** At each architecture's best ratio, does replacing randomly sampled negatives with hard-mined negatives (frames on which a baseline model produces false positives) yield further, architecture-dependent gains?

Our contributions are threefold:

1) A cross-architecture benchmark of negative-frame ratio sensitivity on a real AIoT dataset, testing whether the non-monotonic inverted-U relationship reported in prior single-architecture studies holds consistently across CNN-based and transformer-based lightweight detectors.
2) A hard-negative curation protocol and cost-benefit comparison against random negative sampling at matched dataset size, quantifying whether the additional curation effort is justified and whether the answer differs by architecture.
3) A deployment-cost reframing of false-positive rate: translating measured FP-per-frame differences into estimated nuisance-alert rates per hour, directly connecting an accuracy metric to a consequence AIoT system designers plan around.

II. Related Work

Negative sampling has long been recognized as a first-order design choice in object-detector training. At the loss level, Focal Loss [1] was introduced because the vast number of easy negatives in dense-prediction detectors can overwhelm training; it addresses imbalance through loss reweighting rather than through the ratio of negative instances itself. Related work shows that negatives are not static targets: they can drift toward foreground appearance as training proceeds, confusing the detector's foreground-background boundary and motivating dynamic re-selection [2]. In anchor-based versus anchor-free comparisons, the rule used to assign positive and negative training samples has a larger effect on performance than the underlying regression formulation [3], reinforcing that sample assignment —rather than architecture alone—drives much of a detector's behavior. At the region level, mining strategies that surface visually confusable background chips measurably raise mAP relative to training on positive-only regions [4].

A separate line of work treats negative frames as a dataset-curation lever. Standard guidance for one-stage detectors recommends including roughly 0–10% background-only images to reduce false positives [8]. Direct empirical sweeps show the relationship is non-monotonic: one study varying the proportion from 0% to 40% across two YOLO variants found recall and mAP@50 peaked near 20% before declining, with excess negatives causing over-conservative predictions [5]. In a safety-critical driving context, a far higher proportion was needed: raising background images from 10% to over 60% cut false positives by more than 80% while also reducing false negatives by 25% [6], suggesting the effective operating point is strongly domain-dependent. In human–object interaction detection, scaling the negative-to-positive ratio from roughly 10:1 to 1000:1 produced a consistent performance gain [7], underscoring how strongly the "correct" configuration depends on task structure.

Taken together, this literature establishes that negative-sample configuration reliably shifts a detector's precision-recall operating point, but reported optimal configurations vary by an order of magnitude across tasks and are always evaluated on a *single* detector architecture at a time. No existing study holds negative-frame configuration fixed while varying detector architecture, leaving open whether a ratio or curation strategy tuned for one model transfers to another—a gap directly relevant to AIoT practitioners choosing among architecturally distinct lightweight detectors under a shared data budget.

III. Methodology

A. Dataset

Experiments use an in-cabin driver-monitoring (DMS) dataset comprising 15,723 total frames, of which 3,001 contain annotated driver-cue events (`phone_use`, `drinking`, `yawning`, `hand_over_mouth`) and 12,722 are background-only, reflecting a natural negative prevalence of approximately 81%. All frames are partitioned using a stratified 80/10/10 random split (seed = 42) into disjoint training, validation, and test sets. Validation and test sets each contain 1,572 frames (300 positive, 1,272 negative; 80.9% negative) and are held fixed across all experimental conditions. Evaluating on a natural-distribution test set is deliberate: an artificially rebalanced test set would mask the false-positive behavior that negative-frame configuration is intended to address, and would not reflect the negative-dominant conditions encountered in continuously streaming AIoT deployments.

B. Detector Architectures

Three lightweight detectors, selected to span distinct design paradigms relevant to edge deployment, are trained under identical conditions. **YOLO11n** [9] is an anchor-free, one-stage CNN detector representing the current mainstream YOLO lineage. **YOLO26n** [10] is the succeeding YOLO generation, included to test whether architectural refinements within the same CNN lineage alter sensitivity to negative-frame configuration relative to YOLO11n. **D-FINE-N** [11] is a transformer-based, query-driven real-time detector, included to test whether a fundamentally different detection paradigm (set-based prediction vs. dense anchor-free prediction) responds differently to negative-frame manipulation. All three are evaluated at their smallest ("nano") variant to reflect realistic AIoT compute budgets. Table I provides architecture context; since inference latency is governed by model structure, not training-data composition, it is reported as fixed context rather than as an experimental variable.

C. Negative-Frame Configurations

Ratio Sweep (RQ1). For each architecture, five training sets are constructed by holding the full 2,401-frame positive core fixed and sampling nested negative subsets at exact arithmetic 20%-point steps: 0% (0 neg./2,401 total), 20% (600/3,001), 40% (1,600/4,001), 60% (3,602/6,003), and 80%

TABLE I
DETECTOR ARCHITECTURE OVERVIEW (640×640 INPUT)

Model	Params	FLOPs	Paradigm
YOLO11n	2.6 M	6.5 G	Anchor-free CNN
YOLO26n	2.5 M	6.3 G	Next-gen CNN
D-FINE-N	4.3 M	25.0 G	Transformer (RT-DETR)

Target deployment: NVIDIA Jetson Orin Nano 8 GB.

(9,604/12,005). Negatives are sampled as strictly nested subsets ($\mathrm{Neg}_{0\%} \subset \mathrm{Neg}_{20\%} \subset \cdots \subset \mathrm{Neg}_{80\%}$), yielding $3 \times 5 = 15$ training runs.

Hard-Negative Curation (RQ2). For each architecture, the best-performing ratio from the RQ1 sweep anchors a controlled comparison: random subsampling vs. hard-negative mining at matched dataset size. The mining protocol is: (i) train a baseline on the positive-only split; (ii) run inference over the full 10,178-frame negative candidate pool; (iii) retain frames producing a false-positive detection at confidence $\tau \geq 0.25$ (standard driver-monitoring threshold); (iv) backfill with randomly sampled negatives if the mined count falls below the target. This adds $3 \times 2 = 6$ runs, for **21 total training runs**.

D. Training Protocol

All models are fine-tuned from official pretrained COCO checkpoints at 640×640 resolution under mixed precision (AMP), with a single deterministic seed (seed $=$ 42) across all runs. Table II summarizes the frozen configuration. YOLO11n and YOLO26n use the Ultralytics framework with `optimizer: auto` (weight decay 0.0005, mosaic augmentation disabled for the final 10 epochs). D-FINE-N uses AdamW with official unscaled learning rates (backbone 4×10^{-4}, head/transformer 8×10^{-4}, weight decay 10^{-4}, EMA restart decay 0.9999). For D-FINE-N, 8 GB VRAM constraints require a physical batch size of 4 with 8-step gradient accumulation (effective batch 32); this is explicitly documented as a hardware adaptation, not a claim of numerical equivalence to the official distributed batch size of 128. All non-negative-frame hyperparameters are held constant within each architecture across ratio levels, isolating negative-frame prevalence as the sole experimental variable.

TABLE II
FROZEN TRAINING CONFIGURATION (NVIDIA RTX 4060 8 GB)

Parameter	Y11n	Y26n	D-FINE-N
Physical batch	16	16	4
Effective batch	16	16	32
Epochs	100	100	160
Optimizer	auto	auto	AdamW
Weight decay	0.0005	0.0005	0.0001
AMP	Yes	Yes	Yes
Aug. stop (epoch)	90	90	148

All hyperparameters are held constant across ratio levels within each architecture. Negative-frame ratio is the sole experimental variable.

E. Evaluation Metrics

Each trained model is evaluated on the fixed, natural-distribution test set using two categories of metrics: *(i) Detection accuracy:* Precision, Recall, mAP@50, and mAP@50:95 under standard COCO evaluation protocols. *(ii) Deployment cost:* False positives per 1,000 frames (FP/1k), capturing nuisance-alert frequency under continuous negative dominance, which is the primary operational concern in AIoT streaming applications and drives the cost analysis in Section IV.

Because each configuration is evaluated via a single training run without repeated seeds, all comparisons report point estimates, evaluating practical magnitude and trend directionality without inferential hypothesis testing.

IV. EXPERIMENTAL RESULTS

A. Negative-Frame Ratio Sensitivity (RQ1)

Table III reports detection accuracy and FP/1k across all 15 ratio-sweep runs. The key question is whether the performance-maximizing ratio is the same across all three architectures or is architecture-dependent. Prior single-architecture literature suggests a non-monotonic curve peaking near 20% for generic detection [5] and substantially higher proportions for safety-critical domains [6]; our design provides the first direct cross-architecture comparison.

TABLE III
RATIO SWEEP: DETECTION ACCURACY AND DEPLOYMENT COST (RQ1)

Arch.	Neg.	mAP50	mAP50:95	Prec.	Rec.	FP/1k
YOLO11n	0%	—	—	—	—	—
	20%	—	—	—	—	—
	40%	—	—	—	—	—
	60%	—	—	—	—	—
	80%	—	—	—	—	—
YOLO26n	0%	—	—	—	—	—
	20%	—	—	—	—	—
	40%	—	—	—	—	—
	60%	—	—	—	—	—
	80%	—	—	—	—	—
D-FINE-N	0%	—	—	—	—	—
	20%	—	—	—	—	—
	40%	—	—	—	—	—
	60%	—	—	—	—	—
	80%	—	—	—	—	—

Single-seed (seed = 42); 80.9%-neg. test set; best per arch. in **bold**.

B. Hard-Negative Curation vs. Random Sampling (RQ2)

Table IV compares detection accuracy and FP/1k for random subsampling vs. hard-negative mining at each architecture's best-performing ratio from Section IV-A, at matched dataset size.

C. Operational Deployment Cost Translation

To connect accuracy metrics to real-world AIoT impact, Table V translates FP/1k rates into estimated nuisance-alert

TABLE IV
RANDOM VS. HARD-NEGATIVE MINING AT BEST RATIO (RQ2)

Architecture	Strategy	mAP50	mAP50:95	FP/1k
YOLO11n	Random	—	—	—
	Hard	—	—	—
YOLO26n	Random	—	—	—
	Hard	—	—	—
D-FINE-N	Random	—	—	—
	Hard	—	—	—

'Hard": mined from baseline FP at τ=0.25; matched size.

rates per hour across three representative streaming rates. The translation uses:

$$\mathcal{A}_h = \frac{\mathrm{FP}_{1k}}{1000} \times f_{\mathrm{FPS}} \times 3600 \tag{1}$$

where $\mathcal{A}_h$ is nuisance alerts per hour and f_{FPS} is the streaming frame rate in frames per second. Crucially, inference latency is governed by model architecture, not training-data composition; the FP/1k rate—not latency—determines operational alert frequency and thus deployment usability.

TABLE V
ESTIMATED NUISANCE ALERTS/HOUR: BASELINE VS. BEST RATIO

Architecture	Config	5 FPS	15 FPS	30 FPS
YOLO11n	0% neg.	—	—	—
	Best ratio	—	—	—
YOLO26n	0% neg.	—	—	—
	Best ratio	—	—	—
D-FINE-N	0% neg.	—	—	—
	Best ratio	—	—	—

Computed analytically from FP/1k via (1).

V. DISCUSSION

The central question motivating this work—whether negative-frame configuration can be tuned once and applied across detector architectures, or must be re-tuned per architecture—has direct practical implications for AIoT teams allocating a fixed data-curation budget. Existing literature establishes the importance of negative frames in isolation but provides no cross-architecture transferability evidence; our 21-run experimental design fills this gap directly.

If results confirm architecture-dependent optima: architecture selection and negative-frame curation are not independent decisions. A practitioner who tunes the negative-frame ratio on one architecture and then swaps in a different lightweight detector—a common occurrence as newer edge-optimized models are released—cannot assume the same configuration remains near-optimal and should budget for a re-tuning pass. *Conversely, if a consistent optimum emerges* across YOLO11n, YOLO26n, and D-FINE-N, it suggests that—at least among lightweight nano-scale detectors—the negative-frame ratio can be treated as a property of the dataset rather than the model, substantially simplifying curation pipelines for teams evaluating multiple architectures under a shared budget.

The deployment-cost framing in Section IV reinforces why this distinction matters beyond benchmark accuracy: a difference in FP/1k translates directly into a fold-change in nuisance alerts per hour, a gap invisible in aggregate mAP metrics but directly consequential for driver trust, edge compute wake-ups, and uplink bandwidth in continuously streaming AIoT systems.

Limitations. This study is conducted on a single dataset and domain (in-cabin driver monitoring) which, while representative of negative-dominant AIoT deployment conditions, may not transfer to domains with different background statistics or object density. We test three architectures at their nano variants only; larger variants may exhibit different sensitivity. Each configuration is evaluated via a single training run (single seed), producing point estimates that preclude formal statistical hypothesis testing. The hard-negative mining protocol uses a fixed threshold ($\tau = 0.25$) and a single baseline checkpoint per architecture; more sophisticated online mining strategies remain as future work. Deployment-cost estimates are derived analytically from FP-per-frame rates rather than measured on physical edge hardware, and should be read as directional estimates.

VI. CONCLUSION

This paper presents the first cross-architecture study of negative-frame training configuration for lightweight object detectors in a real-world, safety-critical AIoT setting. Training YOLO11n, YOLO26n, and D-FINE-N on a 15,723-frame driver-monitoring corpus under systematically varied negative-frame ratios and curation strategies, we provide the first empirical evidence on whether the performance-maximizing negative-frame configuration is architecture-invariant or architecture-specific. By translating accuracy differences into estimated nuisance-alert rates per hour, we demonstrate that training-data curation choices carry deployment consequences that are invisible in standard mAP metrics but directly consequential for continuously streaming AIoT systems, while leaving inference latency—governed by architecture alone—unaffected.

These findings directly address a practical question for AIoT practitioners: whether negative-frame curation must be re-tuned when switching among lightweight detector architectures, with consequences for any team maintaining a real-time edge detection pipeline. Future work includes extending this analysis to additional AIoT domains and object categories, testing scalability across model sizes within each architecture family, and validating the deployment-cost estimates against measured power and bandwidth telemetry on physical Jetson edge hardware.

ACKNOWLEDGMENT

The authors thank the University of Sharjah for supporting this research.

References

[1] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2017, pp. 2980–2988.

[2] H. Chen, Y. Wang, G. Wang, and Y. Qiao, “Improving object detection with consistent negative sample mining,” in *Proc. LNCS (ICIG)*, 2019.

[3] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Li, “Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection,” *arXiv:1912.02424*, 2019.

[4] B. Singh, M. Najibi, and L. S. Davis, “SNIPER: Efficient multi-scale training,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2018.

[5] Z. Deng *et al.*, “Floating-debris detection with empirical negative sampling,” *arXiv:2510.23798*, 2025.

[6] T. Rezaei, J. Kolb, and W. Stork, “Deep neural network based roadwork detection for autonomous driving,” *arXiv:2604.02282*, 2026.

[7] T. Gupta, A. Schwing, and D. Hoiem, “No-frills human-object interaction detection,” in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2019.

[8] Ultralytics, “Tips for best training results: Background images,” *Ultralytics Documentation*, 2024. [Online]. Available: https://docs.ultralytics.com/guides/model-training-tips/

[9] Ultralytics, “YOLO11,” *GitHub repository*, 2024. [Online]. Available: https://github.com/ultralytics/ultralytics

[10] Ultralytics, “YOLO12,” *GitHub repository*, 2025. [Online]. Available: https://github.com/ultralytics/ultralytics

[11] Z. Peng, W. Wang, L. Dong, Y. Wei, Y. Huang, and F. Wei, “D-FINE: Redefine regression task in DETRs as fine-grained distribution refinement,” *arXiv:2410.13842*, 2024.